

Renato Dei Marini

Junior Software Engineer — Backend / DevOps / Systems

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PROFILE

Developer trained at the Common Core of **42 Firenze: twelve published projects**, from low-level systems in C and C++ to containerised microservices and Kubernetes. Used to designing from scratch under hard constraints and to verifying what I write instead of taking it on trust. Looking for a first junior position in backend, DevOps or system tooling.

TECHNICAL SKILLS

- **Programming:** C, C++, TypeScript (Node.js), Bash; YAML for Kubernetes manifests, Docker Compose and configuration.
- **Tools:** Git, Make/Makefile, GCC and Clang, valgrind; Docker and Docker Compose (multi-stage Dockerfiles), NGINX, Kubernetes (K3s, k3d, kubectl), Helm, Argo CD, Traefik, Vagrant and VirtualBox; NestJS, Prisma, PostgreSQL, Redis, MariaDB, Swagger, openssl; POSIX threads, POSIX semaphores, BSD sockets, GNU readline, MiniLibX (X11).
- **Backend and infrastructure:** microservices with per-service data ownership, transactional REST APIs, versioned migrations, reverse proxying with TLS, containerisation, infrastructure as code and GitOps.
- **Systems:** concurrency (threads, mutexes, semaphores), I/O multiplexing through one `poll()`, UNIX system programming, interpreters, manual memory management without leaks.

PROJECTS

ft_transcendence

github.com/redei-ma/ft_transcendence

Team project (5 people), role: Project Manager — User Service & Database

- Online 1v1 browser game platform, **microservice** architecture: four **NestJS** applications in TypeScript, **PostgreSQL** and Redis, ten **Docker** containers on a private network behind a single **NGINX** gateway that terminates TLS and is the only one publishing a port.
- Mine: the **entire user-service** — 8 controllers, 9 services, **48 REST endpoints**, notifications over SSE — the **Prisma** schema (11 models, **14 versioned migrations**) shared by four services, with end-of-match writes in a **single transaction**, and the Docker infrastructure (six multi-stage **Dockerfiles**, healthchecks, ordered dependencies), plus shared work on NGINX with TLS and rate limiting.

Inception-of-Things

github.com/redei-ma/Inception-of-Things

Solo project

- Four **Kubernetes** environments of increasing complexity, provisioned from scratch: a two-node **K3s** cluster on **Vagrant** VMs; three apps behind an Ingress routing on the **Host** header; k3d with **Argo CD** deploying from the manifests in the repository; a bonus with GitLab inside the cluster, closing the **GitOps** loop locally.
- Infrastructure as code in Bash and YAML, **prune** and **selfHeal** reconciliation, a **Helm** chart trimmed down through **values.yaml**, components pinned to explicit versions; cross-architecture emulation (amd64 image on an arm64 VM via **qemu-user-static**).

ft_irc

github.com/redei-ma/ft_irc

Team project (3 people)

- An IRC server in **C++98** that talks to real clients: five classes with separated responsibilities, sixteen commands, the five required channel modes. I/O multiplexing through a **single poll()** on non-blocking sockets, in one process, without threads, **fork** or external libraries.
- TCP carries a stream, not messages: a per-client buffer only executes commands terminated by `\r\n`. Silent memory defects (use-after-free, out-of-bounds reads) diagnosed with **valgrind**.

Nine further 42 projects — C, C++ and concurrency, system programming, Docker, graphics: **minishell** (a Bash-like shell: lexing, parsing and execution), **inception** (eight containers from hand-written **Dockerfiles**), **philosophers**, **push_swap**, **pipex**, **libft**, **cpp-modules**, **cub3D**, **so_long** — all on github.com/redei-ma.

EDUCATION

42 Firenze — Post-Common Core specialisation, towards the RNCP qualification 2026 – present

42 Firenze — Intensive software engineering programme 2024 – 2026

Common Core completed. Peer-to-peer method based on hands-on projects and self-directed reading of technical documentation, with strict code reviews.

Istituto Tecnico Commerciale Statale Einaudi-Gramsci, Padua 2012 – 2017

Technical secondary school diploma in Business Administration, Finance and Marketing

Languages — Italian native; English: daily technical reading and writing.

WORK EXPERIENCE (NON-IT)

• **Electrician** — Residential and industrial installations; teamwork and safety regulations. 2019 – 2024

• **Waiter** — High-pressure customer service, multitasking. 2018 – 2019